American Journal of Health Education

ISSN: 1932-5037 (Print) 2168-3751 (Online) Journal homepage: www.tandfonline.com/journals/ujhe20

Exergame Apps and Physical Activity: The Results of the ZOMBIE Trial

Joan Cowdery, Paul Majeske, Rebecca Frank & Devin Brown

To cite this article: Joan Cowdery, Paul Majeske, Rebecca Frank & Devin Brown (2015) Exergame Apps and Physical Activity: The Results of the ZOMBIE Trial, American Journal of Health Education, 46:4, 216-222, DOI: 10.1080/19325037.2015.1043063

To link to this article: https://doi.org/10.1080/19325037.2015.1043063

Published online: 06 Jul 2015.

Submit your article to this journal

Article views: 1933

View related articles

View Crossmark data

Citing articles: 5 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ujhe20

American Journal of Health Education, *46*, 216–222, 2015
Copyright © SHAPE America
ISSN 1932-5037 print/ISSN 2168-3751 online
DOI: 10.1080/19325037.2015.1043063

Routledge
Taylor & Francis Group

Exergame Apps and Physical Activity: The Results of the ZOMBIE Trial

Joan Cowdery, Paul Majeske, and Rebecca Frank
Eastern Michigan University

Devin Brown
University of Michigan

Background: Although there are thousands of health and fitness smartphone apps currently available, little research exists regarding the effects of mobile app technology on physical activity behavior. **Purpose:** The purpose of this study was to test whether Exergame smartphone applications increase physical activity levels. **Methods:** This was a 12-week randomized, controlled, parallel group trial. The intervention consisted of the use of Exergame smartphone apps and motivational messaging. Validated instruments were used to measure physical activity, enjoyment, motivation, and competence. **Results:** Forty subjects were randomized and completed baseline assessments; 39 (97.5%) completed the 12-week follow-up. Median age was 32 years (interquartile range [IQR] = 25, 41.75); 85% were women. No differences between groups were identified in primary or secondary outcomes. Within group, physical activity decreased in the controls and autonomous motivation increased in the intervention group. **Discussion:** Exploratory findings were interesting regarding the use of Exergames to encourage physical activity. The intervention group showed less of a decline in activity, suggesting a possible attenuation of the observed seasonal fluctuation by the use of the Exergames. **Translation to Health Education Practice:** Given their popularity, health educators should continue to explore the use of Exergame apps as a tool to facilitate physical activity.

BACKGROUND

Physical inactivity is a major contributor to several of the leading causes of death in Americans: cardiovascular disease, stroke, diabetes, cancer, and the national epidemic of obesity. Yet, most U.S. adults fall far short of guideline recommendations for physical activity. Recent estimates show that less than 21% of U.S. adults meet guidelines for both muscle-strengthening (29.3%) and aerobic (51.6%) physical activity.[1] This suggests a need for continued efforts in the identification and evaluation of additional behavior change interventions and programs.

As technology has and continues to evolve, health educators have consistently researched and developed new techniques for utilizing emerging technologies in the delivery of health education and behavior change interventions. Approaches as varied as Internet-based programs, computer-tailored interventions, and use of virtual worlds such as Second Life have all been shown to be appealing and effective for behavior change in areas such as physical activity, dietary behavior, and smoking cessation, to name a few.[2-4] Within the last decade, research has focused on the health benefits of interactive video game technology (i.e., Exergames) particularly for encouraging participation in physical activity among both healthy[5-7] and ill or disabled populations.[8-10] Exergames represent a method to increase physical activity by making the activity more appealing and increasing motivation. Traditionally, Exergames have included an interactive video component integrated into an existing piece of exercise equipment (e.g., stationary bike, treadmill). Research on Exergames has consistently shown positive results regarding exercise attendance and adherence,[5] enjoyment,[7] and energy expenditure in both children and adults.[6,7,11]

Submitted January 8, 2015; accepted March 3, 2015.
Correspondence should be addressed to Joan Cowdery, School of Health Promotion and Human Performance, Eastern Michigan University, 318 Porter Building, Ypsilanti, MI 48197. E-mail: jcowdery@emich.edu

Previous research on the effects of exergaming on physical activity in youth has shown mixed results with respect to meeting physical activity guidelines.[12-14] In a review of Exergaming research prior to 2010, Best[12] identified an overall weak association between Exergaming and an increase in physical activity among youth, with a gradual decrease over time as participants become bored with the game. However a systematic review of the use of Exergaming to address childhood obesity demonstrated an overall increase in levels of physical activity as well as a reduction in body fat in 9 studies on children up to 15 years of age.[13] In one of the largest population-based studies of Exergaming and youth ($n = 1209$), results showed that 73% of self-reported Exergamers met moderate to vigorous physical activity guidelines on at least 2 days per week.[14] All of the reported research to date has assessed Exergaming as it is performed with the use of console or equipment-driven games such as Dance Dance Revolution, Wii Fit, and Xbox Kinect.[12-14]

PURPOSE

Though the use of Exergames seems to hold promise for increasing physical activity behaviors, until recently they have been limited by their need for either expensive exercise equipment, game consoles, or TV or computer monitors. Further limiting the application of these games is that they have to be played in a stationary spot where the equipment can be set up. However, with the emergence of mobile smart phone technology, the potential exists for developing behavior change strategies that can be delivered via technology and equipment that many people already possess and that is portable. Current estimates are that there are over 110 million smartphone users in the United States, accounting for 54% of mobile phone use. As of June 2012, 2 out of every 3 new phone purchases is a smartphone.[15] The use of mobile devices for promoting health behavior change is appealing for a multitude of reasons, including the ability of users to access information, for the technology to capture and transmit personal data, and for the ability to program the devices to provide cues to action.[16] Although there are over 18,000 apps currently available in the Health and Fitness section of the Apple App Store,[17] very little research exists regarding the utility and effectiveness of mobile app technology for behavior change.

The purpose of this study, therefore, was to test whether Exergame smartphone applications encourage and increase participation in physical activity, specifically walking and jogging. It was hypothesized that adults randomized to receive the Exergames for 12 weeks would have a greater increase in physical activity compared to a control group. Additional aims included the examination of the impact of the use of Exergame apps on enjoyment of exercise and motivation to exercise.

METHODS

Preliminary Study

The Zombie and Other Mobile Behavior Change Intervention Exergames (Z.O.M.B.I.E.) project consisted of 2 distinct phases. A preliminary qualitative study was conducted in order to identify the Exergame apps to be used in the subsequent Z.O.M.B.I.E. trial. In the preliminary study, a total of 21 participants, recruited by on campus fliers, downloaded 5 fitness apps to their smartphones. The apps were all commercially available for both iPhone and Android platforms and consisted of a variety of tracking, workout planning, goal setting, and entertainment features. The 5 apps evaluated were The Walk, Fitocracy, Nexercise, Runno, and Zombies, Run! Participants were asked to use each app at least twice and then complete an online survey that included the Exergame Experience Rating[18] as well as the Exergame Usability Survey.[19] Subjects were given approximately 6 weeks to use and rate the apps, after which time they were asked to participate in one of 3 focus groups. Overall themes identified during the focus groups were that using an exerapp increased motivation to exercise and increased duration of exercise particularly when using the Exergame apps vs. the more traditional tracking apps. Participants showed a high level of receptivity to the use of fitness apps, both Exergame and tracking type apps, with many individuals having a clear preference for one over the other. Results from the rating surveys and focus groups were used to select the top 2 apps to be used in the subsequent trial.

Trial

We tested the intervention in a 12-week randomized, controlled, parallel-group trial. Adult subjects were identified and recruited through paper and online (including social media) advertisements and word of mouth in and around a university campus in an urban area of southeast Michigan. Participants were deemed to be eligible if they were between the ages of 18 and 69, had regular access to a smartphone with the Android (4.0 or later) or iPhone (IOS 7.0 or later) platform, had no physician-imposed limitations on physical activity, and had not had a myocardial infarction, coronary artery bypass surgery, or coronary stenting procedure within the prior 5 years. The age range for participants was determined a priori based on prior validity studies for the primary outcome measure (see below). The intervention was based on self-determination theory (SDT), a general theory of human motivation that focuses on environmental factors that hinder or facilitate autonomous behaviors. The theory proposes that individuals have innate psychological needs that form the basis for self-motivation and include perceived autonomy and perceived competence.[20] This study was approved by the university's Human Subjects Review Committee.

Assessments

After providing informed consent, participants completed a brief background survey that collected information regarding demographics, typical physical activity, as well as a brief health history. Specific outcome measures included the International Physical Activity Questionnaire (IPAQ)–Short Form, a 7-item tool that has been validated in adults up to age 69, which was used as the primary outcome.[21] The IPAQ is the most commonly used and widely validated physical activity questionnaire.[22] It estimates the continuous measure of total metabolic equivalent (MET)-minutes per week. MET-minutes per week were calculated as $8 \times$ vigorous $+4 \times$ moderate $+3.3 \times$ walking (low). The IPAQ assessment asked participants about the time spent being physically active (vigorous, moderate, walking) in the last 7 days. Secondary outcomes included the Physical Activity Enjoyment Scale, an 18-item tool that has been validated in adults, that was used to assess the extent to which an individual enjoys doing a particular physical activity.[23] Theoretical constructs related to SDT were collected through the use of the Treatment Self-Regulation Questionnaire (TSRQ) for exercise (15 items) and the Perceived Competence for Exercising Regularly scale (4 items).[24] The TSRQ consists of 3 subscales designed to measure autonomous motivation (6 items), which was identified as an intermediate target of the intervention; controlled motivation (6 items); and amotivation (3 items). With regards to physical activity, *autonomous motivation* is defined as an intrinsic motivation to engage in an activity for its own sake and often for the simple pleasure and enjoyment of the activity. *Controlled motivation* is an extrinsic drive to participate in the activity because of an expected outcome, and *amotivation* is a lack of desire or intention to engage in the activity.[25]

Additional baseline measures included blood pressure (the average of 2 measurements when possible) taken using an automated device (Omron-BP-760) and usual techniques,[26] height (Seca mobile stadiometer), and weight (A&D UC-321 scale). Twelve weeks after enrollment, participants were scheduled for a follow-up session during which all baseline assessments were repeated. All data collection occurred in a campus conference room. Participants were provided with a $25 incentive for each data collection session.

Intervention

The Exergame apps that were selected for use in the intervention based on the preliminary study were Zombies, Run! and The Walk. These are both commercially available action adventure games. Zombies, Run! is an immersive running game and audio adventure that instructs players to collect supplies and avoid being attacked by Zombies as they exercise. The Walk is also an audio adventure game that presents episodes and challenges to the player, who is tasked with a package that must be delivered in order to save the world. In order to stay alive, the player must walk/run the length of the United Kingdom.

After completion of the initial assessment, participants were randomly assigned to either the Exergame intervention group or the control group. Participants randomized to the intervention group had one of the 2 Exergame apps placed on their smartphones. Participants were given a choice between the 2 apps, based on appeal of the content. Instructions were provided about their use and participants were instructed to use the apps for the next 12 weeks when they walked or ran for exercise. Participants were also assisted in the download and activation of an activity tracking app (MOVES) utilizing a unique study specific e-mail address and password. Participants in the intervention group also received weekly motivational e-mails. The brief messages were based on SDT and designed to increase intrinsic motivation for physical activity. An example of one of the messages is as follows:

> *Dear Z.O.M.B.I.E. Exerstudy Participant,*
>
> *Hi, we hope you are having a fantastic week. Did you know that the Physical Activity Guidelines for Americans recommend that adults get at least 150 minutes per week of moderate intensity activity? Think about how your activity level compares and remember …*
>
> *The only person you are destined to become is the person you decide to be. —Ralph Waldo Emerson*
>
> *Keep up the good work!*
>
> *—your ZOMBIE Exerstudy research team*

Control group participants did not receive the exergame apps or the motivational e-mails. Participants in the control group were similarly assisted in downloading the activity tracking app (MOVES) to their smartphones. This app continuously monitored their activity (frequency, duration, intensity, and distance) and thus was visible to all participants and allowed the study team to download activity data from the MOVES website. Participants in both groups were instructed to leave the app running and to carry their smartphones with them at all times. Data collected with the MOVES tracking app were examined to determine its feasibility for passive collection of activity data.

Sample Size Calculations, Randomization, and Statistical Analysis

Based on estimates from a study of similar design, it was determined a priori that 17 subjects in each treatment group (intervention and control) would be needed to detect a between-group difference of 100 MET-min per week assuming a difference (12 week vs. baseline) in mean MET-min of 200 in the intervention group and 100 in the treatment group and a standard deviation of 100.[27] Subjects

were randomized to treatment group using blocked randomization, with blocks of 8, using the statistical package R 3.0.2 (R Foundation for Statistical Computing, Vienna, Austria). Intent-to-treat principles were followed for main analyses. The primary analysis compared differences between the intervention and control group between 12-week and baseline IPAQ scores using a Wilcoxon signed-rank test. Other outcomes were compared similarly. Exploratory analyses comparing baseline and 12-week results within groups were also conducted using Wilcoxon signed-rank test for pairs. Between-group differences in MOVES data were explored using Wilcoxon signed-rank tests.

RESULTS

Forty subjects were randomized and completed all baseline assessments between June and August of 2014. Nearly all participants ($n = 39$, 97.5%) completed the 12-week follow-up assessments (conducted September through November 2014). Baseline characteristics of the intervention and control groups are presented in Table 1. Median age was 32 years (interquartile range [IQR] = 25, 41.75). The majority of participants were female (85%); most (85%) were white. The only self-reported cardiovascular risk factor was hypertension ($n = 3$, 7.5%). According to body mass index categorizations, the majority of the participants were overweight or obese (67.5%) and the median body mass index was 26.29 (IQR = 23.6, 30.5). Only 7.5% were current smokers. The majority of participants reported walking (72.5%) or running (48.7%) for exercise at least once per week with 60% reporting that they walked for exercise several times per week. From the baseline IPAQ data, the mean frequency for the week for vigorous days of exercise was 2.6 (SD = 1.7) with a mean duration of 53 minutes (SD = 43). For moderate activity, participants reported a mean weekly frequency of 2.2 (SD = 1.8) days with a mean duration of 56 minutes (SD = 53). Of those who reported walking or running for exercise, over 80% reported typically taking their phone with them. About three quarters (72.5%) reported that they had previously used a fitness phone app.

Primary and Secondary Outcomes

A comparison of the differences between intervention and control groups (12-week minus baseline scores) is provided in Table 2. No differences between groups were identified in the primary outcome, the IPAQ, or the secondary outcomes including the Physical Activity Enjoyment Scale and TSRQ or its subscales when comparing the difference between 12-

TABLE 1
Baseline Characteristics of Study Participants ($n = 40$) by Treatment Group[a]

Demographics, N *(%)*	*Intervention* (n = *20)* (n *(%) or Median (IQR))*	*Comparison* (n = *20)* (n *(%) or Median (IQR))*
Age (years)	31.5 (23.5, 41.8)	32.0 (25.0, 45.5)
Female	18 (90)	16 (80)
Race		
Black	1 (5)	2 (10)
White	19 (95)	15 (75)
Other or multiple races	0	2 (10)
Ethnicity, not Hispanic	19 (95)	17 (85)
Education		
Some college	4 (20)	5 (25)
Bachelor's degree	7 (35)	6 (30)
Advanced degree	9 (45)	9 (45)
Hypertension history	0	3 (15)
Cigarette smoking status—current	3 (15)	0
Walk for exercise		
Daily or several times a week	13 (65)	11 (55)
Take phone when walking—almost all of the time	15 (75)	17 (85)
Run for exercise		
Daily or several times a week	5 (25)	7 (35)
Take phone when running—almost all of the time	9 (45)	11 (55)
Technology proficiency		
Intermediate or advanced	18 (90)	15 (75)
Expert	2 (10)	5 (25)
Exercise or fitness phone app ever used—Yes	15 (75)	14 (70)
Weight (lb.)	168.2 (143.5, 184.9)	171.5 (130.1, 212.5)
Height (in.)	64.5 (63.2, 67.4)	64.0 (63.3, 67.2)

[a]IQR indicates interquartile range.

TABLE 2
Comparison of 12 week-Baseline Differences between Treatment Groups[a]

	Intervention (N = 20)				*Control* (N = 19)				*Between Group (Primary Outcome)*
	Baseline (median (IQR))	*12 week (median (IQR))*	*Difference (median (IQR))*	P *value*	*Baseline (median (IQR))*	*12 week (median (IQR))*	*Difference (median (IQR))*	P *value*	P *value*
Total MET min/week	2039 (1458, 2961)	1809 (817, 2444)	− 356.8 (-1047, 192.4)	0.18	2586 (1542, 4201)	1596 (530, 2148)	− 722 (-1234.8, -122.5)	0.05	0.47
Walk MET min/week	693 (149, 1337)	454 (210, 1386)	− 24.75 (-383.63, 198.00)	1.00	528 (264, 1155)	330 (149, 891)	− 132 (-412.5, 33.0)	0.12	0.32
Moderate MET min/week	240 (20, 1350)	240 (65, 720)	0 (-330, 60)	0.22	240n (180, 660)	120 (0, 300)	− 120 (-240, -30)	0.05	0.57
Vigorous MET min/week	1200 (300, 1776)	480 (0, 1440)	− 180 (-960, 60)	0.20	1080 (460, 2200)	480 (0, 1440)	− 240 (-1000, 120)	0.10	0.87
PACES	99.0 (87.0, 113.8)	105.5 (88.8, 115.0)	.002 (-8.25, 12.25)	0.64	95.0 (82.0, 110.0)	99.0 (85.0, 114.0)	5.00 (-5.50, 14.00)	0.48	0.66
Perceived competence	5.5 (4.4, 6.6)	5.9 (4.6, 6.9)	0.25 (-0.25, 0.81)	0.30	5.50 (5.00, 6.75)	5.8 (3.5, 6.8)	− 0.250 (-0.875, 0.00)	0.09	0.06
Autonomous motivation	35.5 (32.3, 41.0)	39.0 (33.3, 42.0)	1.0 (0.0, 2.5)	0.04	38 (33, 41)	38 (36, 42)	1.0 (0.0, 2.5)	0.14	0.92
Controlled motivation	20.0 (14.5, 25.8)	20.0 (17.3, 25.3)	0.5 (-1.5, 3.0)	0.60	20.0 (16.5, 22.5)	19.0 (14.0, 26.0)	1.0 (-0.5, 4.0)	0.30	0.69
Amotivation	6.0 (4.0, 8.0)	5.5 (3.3, 8.5)	0.0 (-1.25, 0.25)	0.55	5.0 (3.5, 7.5)	6.0 (5.0, 7.0)	1.0 (0.0, 1.0)	0.25	0.16
SBP	107 (101, 112)	110 (104, 114)	4 (-0.25, 8.5)	0.03	115 (105, 124)	121 (103, 125)	2 (-6.75, 7.5)	0.69	0.41
DBP	71 (66, 80)	75 (66, 79)	1 (-0. 5, 4.25)	0.11	75 (71, 80)	78 (70, 82)	1 (-4.0, 5.5)	0.42	0.91
BMI	26.0 (24.6, 30.2)	25.6 (23.4, 29.1)	− 0.0921, (-0.6046, 0.2631)	0.39	27.7 (23.0, 31.3)	28.1 (22.0, 31.9)	0.1879 (-.3494, 0.4557)	0.62	0.37

[a]IQR indicates interquartile range; MET, metabolic equivalent; PACES, Physical Activity Enjoyment Scale; SBP, systolic blood pressure; DBP, diastolic blood pressure; BMI, body mass index.

week and baseline scores. Similarly, no treatment group difference was identified for the levels of physical activity (low, moderate, and vigorous). There was a trend for higher perceived competence in the intervention compared to the control group, although this did not reach significance.

Exploratory Analyses

Within-group differences

Though no differences were identified between groups, within-group changes over time were observed. Total physical activity and moderate levels of physical activity decreased significantly in the control group (Table 2). Autonomous motivation for physical activity increased in the intervention group through time but not in the control group. Additionally, mean systolic blood pressure showed a significant increase in the intervention group.

Device-recorded data

All participants had initially downloaded and activated the MOVES app and were therefore included in the analysis. Median days of data collection was 85 for both intervention (IQR = 84, 86) and control (IQR = 82, 86) groups ($P = .49$). Median minutes of activity per day recorded by the MOVES app were 51.4 (IQR = 32.5, 62.8) for the intervention group and 40.4 (IQR = 31.6, 54.4) for the control group ($P = .46$).

DISCUSSION

As the first known trial designed to investigate the impact of commercially available Exergame apps on physical activity, this study provided several useful insights. Although the trial showed no benefit in its primary and secondary endpoints, some of the exploratory findings were encouraging for the use of Exergames to increase physical activity. The participants in the control group showed a significant decline in physical activity levels over the course of the study that was not observed in the intervention group. This suggests a possible attenuation of the observed seasonal fluctuation in activity level by the use of the Exergame apps and is consistent with previous research on exergaming that typically has shown a modest impact on physical activity levels.[12] The intervention group did show an increase in autonomous (intrinsic) motivation to exercise regularly, perhaps as a result of the use of the Exergames or the addition of motivational messaging. Although not statistically significant, participants in the intervention group also reported higher levels of perceived competence and of overall enjoyment of physical activity consistent with previous research on the use of Exergames.[6,28]

It is also possible that the use of the Exergame apps resulted in an initial increase in activity that was not evident in the IPAQ assessment at week 12 due to the overall decline in activity levels by that point. The lack of a validated objective instrument for the measurement of activity and the reliance on self-report data on activity over the previous 7 days (IPAQ) limited our ability to track activity trends over the length of the study.

The study had several strengths, including the use of a rigorous randomized, controlled design. The study relied on validated outcome measures and was grounded in findings from formative research. Furthermore, for innovative technological tools such as smartphone apps to impact health behavior, they have to be available to the general public. Research to date has been done primarily on apps designed specifically for research and, as a result, are not publicly available. This study aimed to use only commercially available apps both for the Exergames and for the tracking of physical activity. As one of the first trials to be done on the impact of Exergame phone app use on physical activity in adults, this research provides the foundation and reference for future inquiry in this area.

Limitations

Possible explanations for the lack of positive findings include that the intervention was not effective or that the intervention worked but, due to a smaller effect size than was anticipated, the study was not adequately powered. Additionally, the overall time frame for the study was short and there may have been a seasonal effect on overall physical activity among the study sample. The 12-week study time frame extended through summer into fall, which resulted in not only a change in weather but also a change in daily schedules for many of the participants as they transitioned back to an academic schedule. Activity levels understandably decreased as individuals began the fall semester either as staff or students.

It is also possible that the MOVES app provided an active treatment component to the control group that was unintended, particularly for those participants who prefer this type of fitness app. The use of the MOVES tracking app was intended to provide a passive, objective method for collection of activity data; however, its use required participants to carry their phones with them during exercise and compliance with this was not monitored. Furthermore, the MOVES app provided an interactive interface for tracking activity. Many of the participants, including those in the control group, provided unsolicited feedback that they found the app useful and enjoyable. Several participants also reported experiencing technical issues with the use of the app and were frustrated by the amount of data and battery power it required. This may have resulted in participants not allowing the app to run continuously as it was intended.

The ability to generalize findings is limited in that the study sample may not have been representative of the general population given that the participants tended to be younger,

healthy, and already interested in being physically active and in using fitness apps on their smartphones. Only 2 specific Exergame apps were tested, neither of which were designed with a theoretical framework of health behavior change.

TRANSLATION TO HEALTH EDUCATION PRACTICE

As technology continues to evolve, it is imperative that health educators continue to explore meaningful applications of these technologies in their efforts to encourage healthy behavior change. Furthermore, as smartphone use continues to expand, health educators need to understand how people are using their smartphones and their applications to help with changing and maintaining health behaviors such as physical activity. It is apparent that people have a generally high level of receptivity to the use of Exergame and fitness tracking apps. It is therefore incumbent upon health educators to identify or even design apps that will be most effective in assisting individuals in meeting their physical activity goals.

REFERENCES

1. Centers for Disease Control and Prevention. Adult participation in aerobic and muscle strengthening physical activities—United States, 2011. *MMWR Morb Mortal Wkly Rep.* 2013;62:326-330.

2. Strecher V. Internet methods for delivering behavioral and health-related interventions (eHealth). *Annu Rev Clin Psychol.* 2007;3:53-76.

3. Beard L, Wilson K, Morra D, et al. A survey of health-related activities on Second Life. *J Med Internet Res.* 2009;11(2):e17. http://www.jmir.org/2009/2/e17/. Accessed December 30, 2014.

4. Webb T, Joseph J, Yardley L, et al. Using the Internet to promote health behavior change: a systematic review and meta-analysis of the impact of theoretical basis, use of behavior change techniques, and mode of delivery on efficacy. *J Med Internet Res.* 2010;12:e4. http://www.jmir.org/2010/1/e4/. Accessed November 18, 2010.

5. Warburton D, Bredin S, Horita S, et al. The health benefits of interactive video game exercise. *Appl Physiol Nutr Metab.* 2007;32:655-663.

6. Graf DL, Pratt LV, Hester CN, et al. Playing active video games increases energy expenditure in children. *Pediatrics.* 2009;124:534-540.

7. Graves LE, Ridgers ND, Williams K, et al. The physiological cost and enjoyment of Wii Fit in adolescents, young adults, and older adults. *J Phys Act Health.* 2010;7:393-401.

8. Hurkmans HL, Ribbers GM, Streur-Kranenburg MF, et al. Energy expenditure in chronic stroke patients playing Wii Sports: a pilot study. *J Neuroeng Rehabil.* 2011;8:1-7.

9. Hurkmans HL, van den Berg-Emons RJ, Stam HJ. Energy expenditure in adults with cerebral palsy playing Wii Sports. *Arch Phys Med Rehabil.* 2010;91:1577-1581.

10. Shih CH, Chen LC, Shih CT. Assisting people with disabilities to actively improve their collaborative physical activities with Nintendo Wii Balance Boards by controlling environmental stimulation. *Res Dev Disabil.* 2012;33:39-44.

11. Lanningham-Foster L, Foster RC, McCrady SK. Activity promoting games and energy expenditure. *J Pediatr.* 2009;154:819-823.

12. Best JR. Exergaming in youth: effects on physical and cognitive health. *Z Psychol.* 2013;221(2):72-78.

13. Lamboglia CM, Silva VT, Vasconcelos Filho JE, et al. Exergaming as a strategic tool in the fight against childhood obesity: a systematic review. *J Obesity.* 2013;2013. doi:10.1155/2013/438364

14. O'Loughlin EK, Dugas EN, Sabiston CM, et al. Prevalence and correlates of exergaming in youth. *Pediatrics.* 2012;130:806-814.

15. Nielsenwire. Two Thirds of new mobile buyers now opting for smartphones. http://blog.nielsen.com/nielsenwire/online_mobile/two-thirds-of-new-mobile-buyersnowopting-for-smartphones/. Published 2012. Accessed November 8, 2012.

16. West JH. There's an app for that: content analysis of paid health and fitness apps. *J Med Internet Res.* 2012;14(3):e72.

17. Pocket Gamer. *App Store Metrics*. http://148apps.biz/app-storemetrics/?mpage=catcount. Published 2012.Accessed November 8, 2012.

18. The Exergame Network. *Exergame Experience Rating System (EERS)*. http://www.exergamenetwork.org/. Published 2012. Accessed December 30, 2014.

19. Ryu YS, Smith-Jackson TL. Reliability and validity of the Mobile Phone Usability Questionnaire (MPUQ). *J Usability Stud.* 2006;2:39-53.

20. Ryan RM, Patrick H, Deci EL, Williams GC. Facilitating health behavior change and its maintenance: interventions based on self-determination theory. *Eur Health Psychol.* 2008;10:2-5.

21. Craig CL, Marshall AL, Sjöström M, et al. International Physical Activity Questionnaire: 12-country reliability and validity. *Med Sci Sports Exerc.* 2003;35:1381-1395.

22. van Poppel MNM, Chinapaw MJM, Mokkink LB, et al. Physical activity questionnaires for adults: a systematic review of measurement properties. *Sports Med.* 2010;40:565-600.

23. Kendzierski D, DeCarlo KJ. Physical Activity Enjoyment Scale: two validation studies. *J Sport Exerc Psychol.* 1991;13:50-64.

24. Williams GC, Ryan RM, Deci EL, *Health-Care Self-Determination Theory Questionnaire Packet*. http://www.selfdeterminationtheory.org. Published 2004. Accessed November 8, 2012.

25. Ryan RM, Deci EL. An overview of self-determination theory: an organismic-dialectical perspective. In: Deci EL, Ryan RM, eds. *Handbook of Self-Determination Research*. Rochester, NY: The University of Rochester Press; 2002;p.e33.

26. Pickering TG, Hall JE, Appel LJ, et al. Recommendations for blood pressure measurement in humans and experimental animals: part 1: blood pressure measurement in humans: a statement for professionals from the Subcommittee of Professional and Public Education of the American Heart Association Council on High Blood Pressure Research. *Hypertension.* 2005;45:142-161.

27. Rome A, Persson U, Ekdahl C, et al. Physical activity on prescription (PAP): costs and consequences of a randomized, controlled trial in primary healthcare. *Scand J Prim Health Care.* 2009;27:216-222.

28. Yang C, Wickert Z, Roedel S, et al. Time spent in MVPA during exergaming with Xbox Kinect in sedentary college students. *Int J Exerc Sci.* 2014;7286-7294.